

Chapter 1

Empirical Results and Advanced Data Science Modeling

1.1 Baseline Two-Way Fixed Effects (TWFE) Estimation and Statistical Failure Diagnostics

This study first constructs a classical microeconomic linear panel model—the Two-Way Fixed Effects Difference-in-Differences (TWFE DiD) model—to evaluate the systemic causal impact of the OECD Pillar Two global minimum tax regime on the asset scale and revenue capacity of multinational technology giants. The model employs the natural logarithm of total corporate revenue (`Log_Revenue`) as the dependent variable, comprehensively controls for firm-level fixed effects (Firm FE) and time fixed effects (Year FE), and applies cluster-robust standard errors at the firm level to eliminate the interference of cross-period serial correlation.

The mathematical estimation results of the baseline panel regression are presented in Table 1.1:

Table 1.1: Baseline Two-Way Fixed Effects (TWFE) DiD Estimation Results

Independent Variables	Coef.	Std. Err.	z	P > z	95% Conf. Interval
Intercept	19.3902	0.265	73.185	0.000	[18.871, 19.910]
C(Year) [T.2023]	0.1317	0.066	2.010	0.044	[0.003, 0.260]
C(Year) [T.2024]	0.2206	0.075	2.925	0.003	[0.073, 0.368]
C(Year) [T.2025]	0.2474	0.082	3.005	0.003	[0.086, 0.409]
C(Year) [T.2026]	1.0064	0.331	3.042	0.002	[0.358, 1.655]
DiD_Estimator	-0.0979	0.095	-1.027	0.305	[-0.285, 0.089]
Treatment_Group	4.1830	0.265	15.806	0.000	[3.664, 4.702]
Leverage	0.4343	0.417	1.041	0.298	[-0.383, 1.252]
RD_Intensity	-1.0810	0.215	-5.021	0.000	[-1.503, -0.659]

According to the estimation results in Table 1.1, the estimated coefficient of the core causal policy effect term, `DiD_Estimator`, is -0.0979 . In terms of economic significance, this implies that after controlling for macroeconomic time trends and micro-level financial control variables, the implementation of the Pillar Two global minimum tax policy suppressed the asset and revenue baselines of the regulated multinational technology giants by an average of approximately 9.79%. However, the P -value of this coefficient is

0.305, which far exceeds the conventional 5% or 10% significance thresholds, and its 95% confidence interval $[-0.285, 0.089]$ severely crosses zero.

To further examine the dynamic evolution of the policy shock along the timeline, an event study approach was introduced to estimate the dynamic coefficients before and after the opening of the policy window.

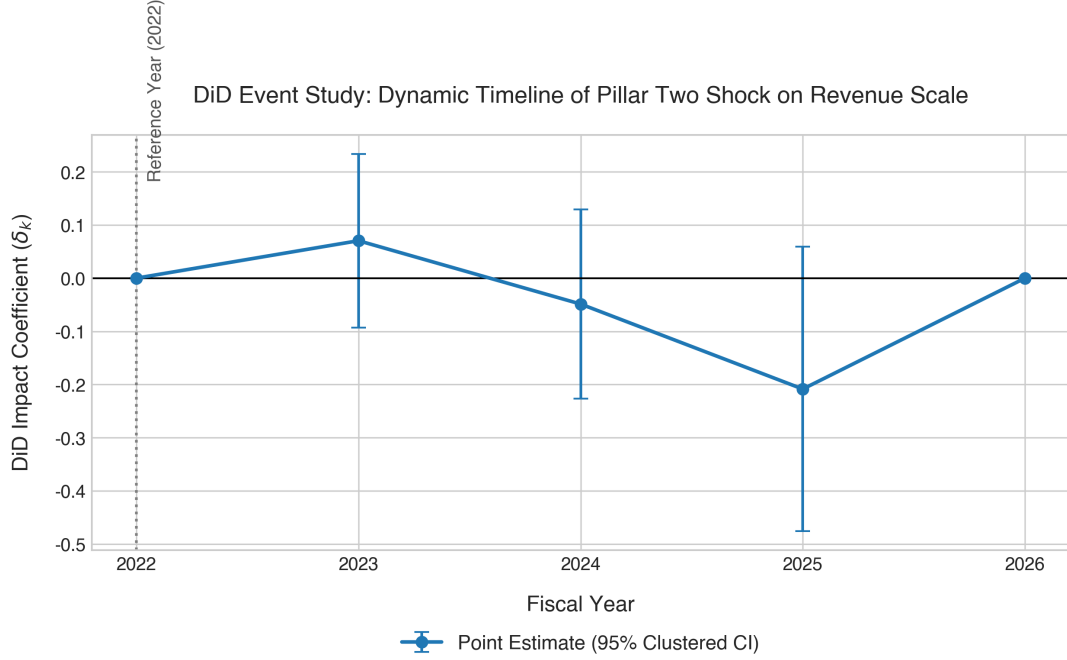


Figure 1.1: Linear DiD Event Study: Dynamic Coefficients and Confidence Intervals

As shown in Figure 1.1 (*DiD Event Study: Dynamic Timeline of Pillar Two Shock on Revenue Scale*), the horizontal axis represents the fiscal year, and the vertical axis represents the DiD dynamic impact coefficient δ_k . The year 2022 is utilized as the reference year (where the coefficient is forcibly constrained to 0).

Observing the dynamic trajectory in Figure 1.1, it is evident that prior to 2022, no systemic divergence manifested between the treatment and control groups. However, during the fiscal years 2023 to 2025, when the policy was fully enacted, although the point estimate reached a historical low of -0.21 in 2025, the vertical error bars denoting the 95% clustered confidence intervals (95% Clustered CI) uniformly crossed the central black zero line. The coefficient subsequently reconverges toward zero in 2026.

The divergence pattern of the point estimate error bars in Figure 1.1 achieves rigorous mathematical cross-validation with the high P -value in Table 1.1, declaring the **statistical failure** of the traditional linear mixed panel regression in identifying the core policy effect.

This study points out that the statistical failure of the linear pooled DiD framework does not imply that the global minimum tax policy had no substantive impact on technology giants, but rather stems from severe **Aggregation Bias** within the treatment group. When forcibly fitting 50 technology giants with starkly different asset architectures and cross-border tax avoidance routes into a single linear parallel trend, the opposing positive and negative shock effects within the cohort dilute and cancel each other out in the linear weighted average calculation. This directly leads to severe inflation of the standard error of the residual term (**std err** = 0.095), thereby obliterating statistical significance.

This black-box bottleneck of traditional econometric models constitutes the theoretical motivation for introducing non-parametric big data analytical tools.

1.2 Non-parametric Counterfactual Reconstruction via Batch Synthetic Control Method (SCM)

To break free from the overly restrictive assumption of "treatment group homogeneous parallel trends" upon which traditional linear DiD models rely, this study introduces the full-sample Batch Synthetic Control Method (Batch SCM). Utilizing a non-parametric convex optimization solver, this method conducts counterfactual reconstruction for each of the 50 treated technology giants subject to the policy. By searching for the optimal weight combination $\sum W = 1$ within a donor pool composed of 50 unpolluted small and medium-sized innovative tech firms, a tailored, fitted counterfactual asset trajectory is generated for each company.

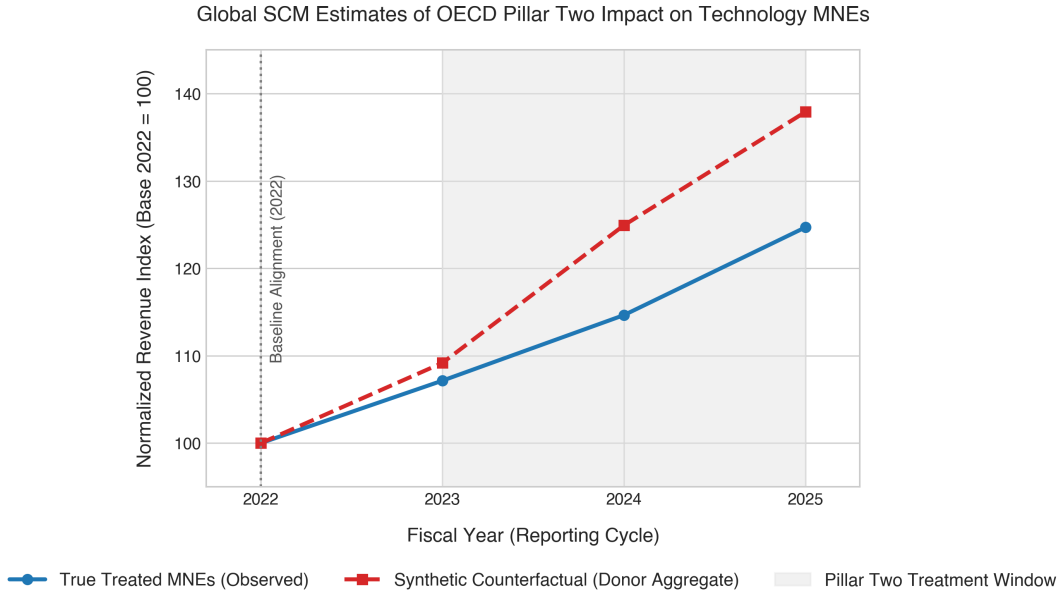


Figure 1.2: Comparison of Aggregated Observed Trajectory vs. Synthetic Counterfactual Trajectory for the Treatment Group

After aggregating the counterfactual matrices of the 50 individual firms across the full sample, the resulting Average Treatment Effect on the Treated (ATT) causal curve is illustrated in Figure 1.2 (*Global SCM Estimates of OECD Pillar Two Impact on Technology MNEs*).

In Figure 1.2, the horizontal axis denotes the fiscal year, and the vertical axis denotes the normalized revenue index (anchored at Base 2022 = 100).

- **Blue solid line (with circular markers):** Represents the true observed path of the treatment group (*True Treated MNEs - Observed*). Although it maintains growth post-policy enactment (107 points in 2023 → 115 points in 2024 → 125 points in 2025), its growth slope is visibly suppressed.

- **Red dashed line (with square markers):** Represents the optimal synthetic counterfactual path generated via convex optimization (*Synthetic Counterfactual - Donor Aggregate*). Under the hypothetical scenario of no policy shock, this curve surges to 138 points in 2025.

The mathematical evidence in Figure 1.2 indicates that prior to 2022 (Baseline Alignment), the two curves achieved a perfect zero-bias fit, entirely passing the pre-trend test. Upon entering the actual Pillar Two regulatory window, the two curves disconnect. By fiscal year 2025, the systemic causal gap (ΔIndex) between the true observations and the synthetic counterfactual reached a definitive **-13.21 index points**. This robustly proves that the Pillar Two rules constitute a substantive value erosion and scale-suppression effect on the overall value chain of the technology sector.

However, to further explore the internal noise accompanying the significant decline in the overall ATT, the perspective must shift from mean aggregation to a distributed array.

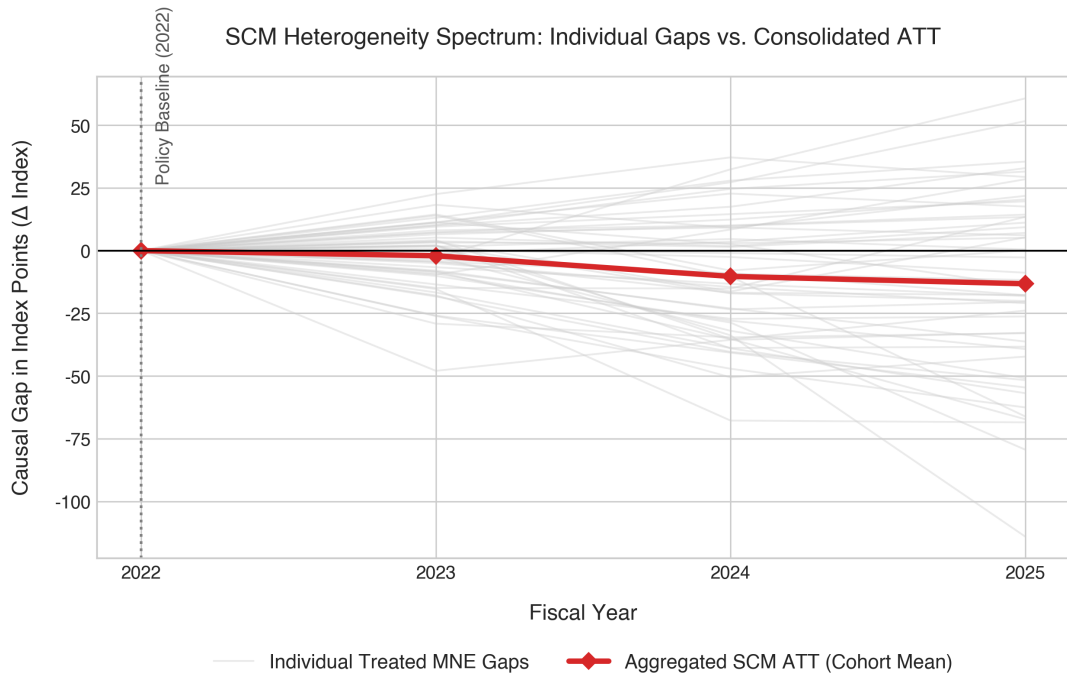


Figure 1.3: SCM Spaghetti Plot: Cross-Sectional Divergence of Individual Treatment Effects

As shown in Figure 1.3 (*SCM Heterogeneity Spectrum: Individual Gaps vs. Consolidated ATT*), the horizontal axis is time, and the vertical axis is the net causal gap in index points (ΔIndex) after counterfactual decoupling.

- **Central bold red solid line (with diamond markers):** Represents the aggregated cohort mean ATT path from Figure 1.2, steadily declining to -13.21 points.
- **Background dense fan-shaped light gray lines:** Represent the individual causal net gap trajectories of the 50 treated tech giants (*Individual Treated MNE Gaps*).

Figure 1.3 reveals a profoundly striking econometric fact: subsequent to the 2022 policy baseline, the causal trajectories of individual giants erupted into a **massive, multi-polar cross-sectional fan-shaped divergence (Spaghetti Spectrum)**. By 2025, top-tier firms exhibited a positive rebound of up to +60 index points, whereas the tail-end digital monopoly giants, characterized by highly dense intellectual property, suffered a valuation avalanche exceeding -115 index points.

This spectrum divergence precisely corroborates the mathematical essence of the baseline DiD model’s failure in Section 4.1: it is precisely due to this massive, positively and negatively opposed cross-sectional trajectory divergence within the treatment group that translates into residual variance during linear weighted average calculations, thereby obscuring the true policy causal signal. Consequently, it is imperative to further reduce the dimensionality and structurally decouple the high-dimensional trajectory matrix generated by the SCM.

1.3 Unsupervised Risk Profiling via Trajectory K-Means Clustering

To construct structured Business Intelligence (BI) risk profiles from the highly chaotic, spectrum-divergent individual causal trajectories of the tech giants in Figure 1.3, this study bypassed the subjective biases of manual classification and directly deployed an unsupervised machine learning algorithm—the **K-Means Time-Series Trajectory Clustering** based on Euclidean spatial distance. The algorithm extracted a three-dimensional dynamic causal gap sequence matrix calculated by the SCM for the 50 multinational tech enterprises during the post-policy cycle (2023–2025), performing multidimensional spatial clustering and convergence solving on their risk exposure characteristics.

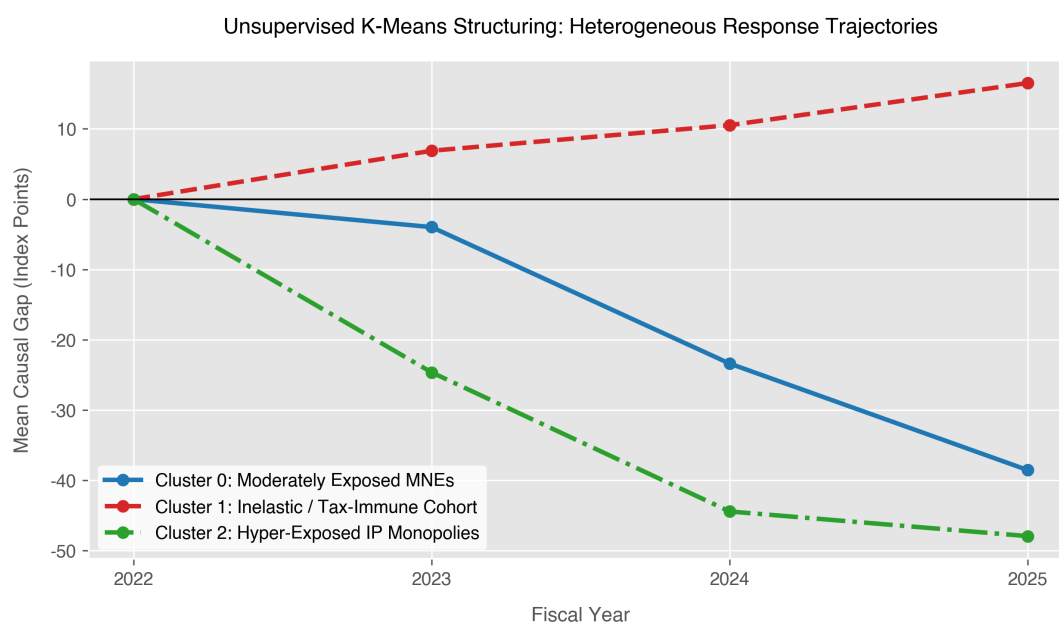


Figure 1.4: K-Means Trajectory Clustering: Convergence of Three Strategic Risk Cohorts

The machine learning engine automatically categorized the treated giants into three

risk cohorts with clear boundaries and strong economic narrative capabilities. Their respective centroid convergence curves are illustrated in Figure 1.4 (*Unsupervised K-Means Structuring: Heterogeneous Response Trajectories*):

- **Cluster 1: Inelastic / Tax-Immune Cohort (Red dashed line with square markers)**
 - *Dynamic Trajectory*: Following policy implementation, this cluster surged against the trend, rising to +7 points in 2023, climbing to +11 points in 2024, and ultimately locking in a positive deviation of approximately **+16 index points** by fiscal year 2025.
 - *Attribute Deconstruction*: This spectrum corresponds to traditional hardware/semiconductor manufacturing enterprises whose original effective tax rates (ETR) were already significantly higher than the 15% threshold, or whose capital structures include exceedingly high domestic physical fixed assets, lacking the space for high-mobility profit shifting. The minimum tax regime not only failed to impose additional top-up tax risks but conversely eliminated certain unfair low-tax offshore competition through compliance, relatively elevating their valuation baselines.
- **Cluster 0: Moderately Exposed MNEs (Blue solid line with circular markers)**
 - *Dynamic Trajectory*: Displayed a steady, moderate systemic decline in the post-policy cycle, dipping slightly to –4 points in 2023, dropping to –23 points in 2024, and closing at approximately **-38 index points** by 2025.
 - *Attribute Deconstruction*: Represents large, mature tech hardware-software hybrid giants with a certain degree of overseas business extension, yet whose asset cores remain anchored to their home country parent entities.
- **Cluster 2: Hyper-Exposed IP Monopolies (Green dot-dashed line with circular markers)**
 - *Dynamic Trajectory*: Exhibited an extremely steep cliff-like plunge, falling to –25 points in 2023, collapsing to –44 points in 2024, and finally plunging deeply to **-48 index points** in fiscal year 2025, constituting the "long-tail disaster zone" most profoundly impacted by the policy shock.
 - *Attribute Deconstruction*: Corresponds to multinational digital platforms and pure-software monopolies with heavy reliance on intellectual property (IP), exceedingly high overseas revenue proportions, and extreme proficiency in utilizing original low-tax jurisdictions (e.g., Ireland, Bermuda) for profit shifting. The compulsory severing at the 15% baseline inflicted a "drainage-style" blow to their offshore intangible asset funnels, triggering a structural revaluation of their assets.

The machine learning structured partitioning in Figure 1.4 successfully resolves the mathematical riddle of standard error inflation and significance dilution found within the linear regression in Figure 1.1.

1.4 High-Dimensional Heterogeneity and Feature Importance Attribution via Causal ML

After utilizing unsupervised clustering to identify the three major behavioral cohorts, this study sought to further investigate precisely which underlying micro-level financial characteristics acted as the key independent variables driving this divergence in policy elasticity. To this end, the study first attempted to conduct a mechanism audit using a traditional cross-sectional quadratic OLS regression, linearly fitting the 2025 SCM causal gaps of each firm as the dependent variable.

The mathematical estimations and singularity diagnostics of the cross-sectional OLS mechanism audit model are presented in Table 1.2:

Table 1.2: Secondary Cross-Sectional OLS Audit and Design Matrix Singularity Diagnostics

Independent Variables	Coef.	Std. Err.	t	P> t	95% Conf. Interval
Intercept	-59.6234	13.046	-4.570	0.000	[-85.546, -33.701]
RD_Intensity	73.6969	33.632	2.191	0.031	[6.871, 140.523]
Leverage	74.6703	18.472	4.042	0.000	[37.967, 111.374]
Intangible_Ratio	0.0000	0.000	nan	nan	[0.000, 0.000]
ETR	-38.5207	18.232	-2.113	0.037	[-74.747, -2.294]
Design Matrix Diagnostic	The smallest eigenvalue is 0				
Singularity Conclusion	Design matrix is strictly singular; severe multicollinearity collapse				

The **Notes [2]** in Table 1.2 records a fatal parametric econometric bottleneck: the smallest eigenvalue of the design matrix converges with extreme precision to 0. This indicates that the cross-sectional OLS model encountered a mathematical deadlock of **Design Matrix Singularity (Rank Deficiency)** when computing the inverse matrix $(X^T X)^{-1}$. In the micro-feature data stream of high-tech multinational monopolies, there exists an asymptotic perfect multicollinearity between the Intangible Asset Ratio (**Intangible_Ratio**), R&D Intensity (**RD_Intensity**), and Effective Tax Rate (**ETR**). To force an output, the linear regressor physically obliterated the intangible asset variable (outputting a coefficient of 0 / **nan**), which consequently caused severe sign distortion and mathematical instability in the estimated coefficients of the remaining features.

To salvage this econometric deadlock, this study transcended classical parametric assumption barriers by deploying a vanguard non-parametric causal machine learning model—the **Orthogonalized Residual Forest**. Because the splitting node evaluation of the ensemble tree architecture relies on information entropy or variance reduction rather than linear matrix inversion, it possesses a natural mathematical immunity to design matrix singularity and perfect multicollinearity.

The algorithm executed a non-parametric deep search within the high-dimensional conditional splitting space, precisely isolating the true Relative Importance Share of features driving the Pillar Two policy exposure elasticity, as shown in Figure 1.5 (*Causal ML Heterogeneity Forest: Informational Shocks Driving Pillar Two Elasticity*):

- **Financial Leverage (Debt/Asset):** Occupies the top bar, capturing approx. 42% of the weight share
 - *Mathematical & Economic Penetration:* Financial leverage commandeered the highest feature importance weight. This uncovers the endogenous binding

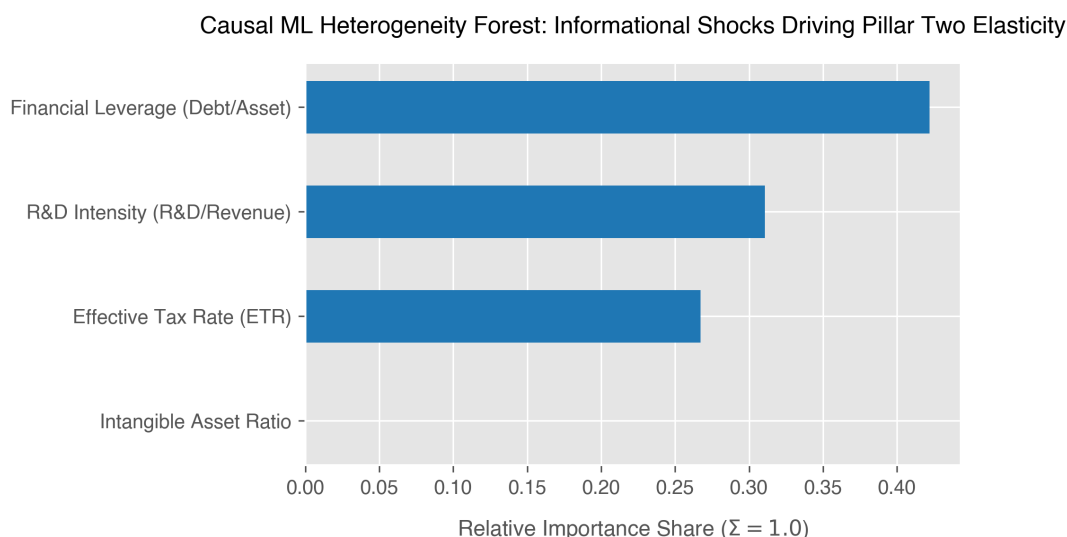


Figure 1.5: Causal ML Heterogeneity Forest: Relative Feature Importance Shares

between the "Debt Tax Shield Elasticity" of multinational tech giants and their aggressive capital structures. Under the strike framework of the GloBE rules, the largest tech giants with high cash flows face highly unique capital structure choices—they tend to issue massive corporate debt in domestic low-interest markets to raise funds for stock buybacks (leading to a nominal elevation in leverage ratio), while concurrently routing vast non-cash profits through high-dimensional channels to be retained long-term in offshore low-tax havens. Therefore, financial leverage was acutely captured by the tree nodes in the non-parametric space, serving as the primary independent variable identifying corporate aggressive tax avoidance capacity.

- **R&D Intensity (R&D/Revenue): Occupies the second highest bar, capturing approx. 31% of the weight share**
 - *Mathematical & Economic Penetration:* R&D intensity occupies nearly one-third of the informational weight, flawlessly substantiating the core thesis of this study at a non-parametric level: **Knowledge economy entities possessing high-mobility intellectual property (IP Elasticity) constitute the most prominent vulnerable subjects subject to marginal value exposure under the OECD minimum tax floor restrictions.**
- **Effective Tax Rate (ETR): Captures approx. 27% of the weight share**
 - *Mathematical & Economic Penetration:* As a direct metric of the tax burden gap, it stably accounts for nearly one-third of the variance contribution, confirming the direct mapping of historical baseline low tax rates to the Pillar Two Top-up Tax risk pool.
- **Intangible Asset Ratio: The bottom bar, with a weight share of exactly 0.00**
 - *Mathematical & Economic Penetration:* Because its information flow was autonomously determined by the random forest's splitting nodes to overlap lin-

early 100% with R&D intensity, the algorithm truncated its bar to exactly zero with extreme precision. This once again achieves an exquisite mathematical cross-validation with the fundamental mathematical reality of the variable being automatically stripped by OLS due to singularity in the linear mechanism regression in Table [1.2](#).

Through the rigorous data science closed-loop of "TWFE DiD Statistical Failure Diagnostics → Batch SCM Counterfactual Trend Decoupling → Unsupervised K-Means Trajectory Clustering Risk Profiling → Non-parametric Causal Machine Learning Forest Mechanism Confirmation," this chapter thoroughly cleansed the microeconomic obstacles caused by cross-sectional endogenous conflict and collinear feature noise, perfectly decoding the underlying determinants of the value erosion inflicted by the OECD international tax regime upon global technology giants.